

Final Year Design Project Proposal

Tracefy: Generating sketches of perpetrators for efficient identification

Names of Team Members Advisor:

Mahad Mohtashim

Mehar Ali Ahmed

Abdul Rafay Khan

Advisor: Dr. Rafia Mumtaz

Co-Advisor: Dr. Muhammad Daud Abdullah Asif

School of Electrical Engineering and Computer Science National University of
Sciences and Technology

Islamabad, Pakistan (2024-25)

Table of Contents

Table of Contents	1
Abstract.....	2
Introduction	2
Problem Statement	3
Literature Review/Existing Products	3
Research Paper 1:	3
Research Paper 2:	7
Research Paper 3:	9
Research Paper 4:	12
Existing System:	15
Project Overview	16
Key components of the project include:	17
Project Development Methodology	17
Implementation Tools and Libraries:	18
Relevance to UN Sustainable Development Goals	18
Work Division	19
Timeline.....	19
Costing.....	20
References:	20
Appendix:	22

Abstract

This document serves as a proposal for Tracefy, an advanced AI-based sketch generation system designed to automatically produce detailed and accurate facial sketches from user inputs. The proposal includes a comprehensive literature review that examines existing methodologies in AI-driven forensic sketch generation, highlighting both their advancements and limitations. By leveraging cutting-edge generative AI models such as Latent Diffusion Models (LDMs) and Generative Adversarial Networks (GANs), Tracefy builds on frameworks like Attribute2Sketch2Face to transform visual descriptions into high-quality facial representations. The system integrates fairness and bias reduction techniques, including Adversarial Debiasing and Regularization Penalties, to mitigate unintended biases related to race, gender, or age. Additionally, the proposal outlines the project's working timeline, costing, and work division, ensuring a structured approach to development and implementation. Tracefy aims to provide law enforcement agencies with a tool that enhances both the accuracy and efficiency of suspect identification while upholding fairness across diverse populations. The proposal also details how Tracefy aligns with key United Nations Sustainable Development Goals (UN SDGs), particularly those promoting peace, justice, and strong institutions, and is committed to improving the effectiveness of crime-solving processes in Pakistan.

Introduction

Criminal activities pose a significant threat to global peace and security hence alerting countries to take necessary steps to control the rise of illegal activities which include, but are not limited to, violent crimes such as killing a person or property crimes like robbery. As of 2023, 83% of the global population lives in countries with high criminality (Global Organized Crime Index). Pakistan, a developing country with the world's fifth largest population, is no alien to urban crimes springing up in its metropolitan cities. Various factors like high inflation, rise in unemployment, regular food shortages, weather calamities and political instability have pushed a large portion of youth below the poverty line, who often end up in the dark underworlds of crime. A bleak picture has been reported by the police in Karachi stating a figure of 60,580 incidents of street crimes, out of which 33.33% belong to episodes of mobile snatching, in the first 9 months of 2022 alone (Ayaz, 2023). In the wake of these rising crime statistics, there is a need to devise solutions to mitigate the threat and work towards enabling a peaceful environment for the residents of the country.

Problem Statement

Criminal sketches drawn by humans are slow, inaccurate and riddled with bias, leading to low apprehension rates"

The application will aim to solve this problem as it will provide speed, accuracy and impartiality. With the fulfilment of these factors, this implementation can be catered to the government. As a B2G category product, it will fill the lack of modern tools and increase apprehension rates, thereby decreasing crime.

Literature Review/Existing Products

The literature review for this project is based on the existing work related to the idea of our project and details about the implementation that we expect to follow. These papers provide useful insight on what types of datasets we can use, and the limitations and advantages of the techniques used in generative-AI technology.

Research Paper 1:

Title: Face Synthesis from Visual Attributes via Sketch using Conditional VAEs and GANs (Di & Patel, 2017)

Authors: Xing Di, Vishal M. Patel

Main Objective:

The main objective of this research paper is to demonstrate the effectiveness of an innovative framework called Attribute2Sketch2Face, which produces facial images from visual attributes. As suggested by the name, this framework is a stage-wise approach based on the combination of Generative Adversarial Networks (GANs) and Conditional Variational Autoencoder (CVAE) which are two significant deep learning models in the field of generative AI. Furthermore, the article explains the methodology in detail along with an overview of the different utilities that were used in the implementation of the project. It also highlights the need for this technology in revolutionizing the process of identifying suspects involved in a crime scene. Finally, the paper compares the results of the proposed framework with the existing methods using performance metrics, concluding that the proposed method is superior in terms of accuracy.

Overview of the Methodology

The framework consists of three stages: attribute-to-sketch (A2S), sketch-to-sketch (S2S), and sketch-to-face (S2F).

Stage 1: Attribute2Sketch

In stage 1, an approximate sketch image is generated from visual attributes. The A2S stage employs a CVAE model which consists of an encoder and a decoder network. The encoder network takes original sketch, visual attributes and noise (random values used to introduce diversity in the output sketch) as inputs and compresses them into a latent vector,

capturing the essential features and patterns of the input data. The latent vector and the attribute vector are then passed into the decoder network. The decoder network learns the relationship between the characteristics of the latent space and attribute vector to synthesize a final sketch. In the testing phase, description of the visual attributes is provided to the A2S model to output sketch of the intended person. The accuracy is tested by comparing the synthesized sketch with the original sketch of the person. Figure 1 shows the results of an A2S model. The odd columns are the results from A2S stage whereas the even columns are the corresponding real sketch images.

Stage2: Sketch2Sketch:

As seen in the Stage 1 results, the synthesized sketches are of poor quality. This is where the second stage, sketch-to-sketch (S2S), comes into action. At Stage 2, the blurred sketches are transformed into sharper sketches with the help of conditional GAN-based architecture which consists of a generator sub-network and a discriminator sub-network. The generator takes blurry sketches as inputs and attempts to generate sharper images, while its adversary, the discriminator, distinguishes between the real and the synthesized sketch, hence improving the quality of the sketch. The generator, G2, makes use of UNet skip connections and DenseNet's dense blocks to refine the sketch as the former retains the low-level details from the blurry sketches while the latter facilitates efficient information flow between the convolutional layers. On the other hand, the discriminator D2, iteratively trains the generator by providing feedback such as pointing out any flaw in the sketches generated by G2. Figure 2 illustrates the steps involved. In this way, convincing sketches are generated which are sharper and more realistic, allowing them to be passed on to the last stage of the framework.

Stage3: Sketch2Face

In the final stage, a colored image is produced from the clear sketch obtained from Stage 2 using similar GAN-based approach consisting of a UNet-based generator and an iteratively training discriminator. The generator makes use of a two-layer ResNet block to integrate the color and texture attributes to produce realistic and detailed colour images. Meanwhile, the discriminator is trained iteratively by pointing out the discrepancies to increase the realness of the generated images.

Together, these three stages demonstrate a comprehensive approach to generate realistic and detailed face images from visual attributes via sketches.

Implementation and Analysis of Results:

The model is trained using three publicly available datasets: CelebA and LFW which contains face images and the corresponding attributes while the CUHK dataset consist of face-sketch image pairs. The training datasets are preprocessed and gradually passed through the above-mentioned stages to train the model. The results from all three datasets are compared with the existing models, namely text2img, StackGAN, CVAE and attr2face as shown in Figure 3. Evaluation is based on the performance metrics such as Inception

Score and Attribute L_2 . The Inception score evaluates the realism and diversity while the Attribute L_2 compares the quality of attributes corresponding to different images, with higher values of Inception Score and lower values of the Attribute L_2 indicating better performance. As shown in the quantitative analysis table in Figure 4, Attribute2Sketch2Face shows the best performance for all the input datasets proving its efficiency and accuracy as discussed by the authors.

Innovation and Contribution

As described by the authors, Attribute2Sketch2Face is a novel process since it divides the problem into stages hence providing a practical and interpretable method for face image reconstruction from abstract descriptions. The problem of converting textual information to visual representation is a complex task posing significant challenges. One of the major challenges is generating high quality images from abstract attributes as there are a vast range of facial features with numerous variations however the proposed method makes the generation process more manageable with the help of intermediate sketches which are less complex to generate but retain the structural information before being converted into a final image (Rosenberg & Ahmed, 2023). In addition, training models to directly generate high resolution images is computationally more intensive whereas a two-stage approach simplifies the process by first focusing on sketch generation which is less resource-intensive. Further, intermediate sketches handle ambiguity in descriptions far more efficiently by focusing on essential features in the first stage and then refining them in the second stage. In summation, this stage-wise approach is a valuable contribution in the field of generative AI technology as it can generate high quality images and achieve significant improvements over other state-of-the-art models.

Relevance to Our Project (Tracefy)

The approach discussed in the paper closely aligns with the concept of our project, especially in its use of generative AI models and its practical implementation for solving real-world problems.

While the framework discussed in the paper introduced us to CVAE, we were already convinced that our model will be primarily based on GANs hence this paper provides useful insights into the use of GANs for generating facial images from visual attributes, especially explaining the integration of UNet and DenseNet architectures into the generator network. Both UNet and DenseNet are highly recommended for image segmentation and classification tasks (Arrorra, 2019), which are integral components of our project.

Moreover, the paper's unique idea of generating sketches from visual features is parallel to our project's goals and serves as a method to enhance our solution. For example, we can leverage the simplicity of sketches to handle various features effectively before progressing to a more detailed image generation. In addition, we can enhance the optimization of our model by breaking down the problem into manageable stages as discussed in the paper. As

a result, we will be able to achieve more accurate results while saving on computational time and resources which is a win-win situation for our project.

Lastly, the paper discusses the benefits of incorporating the model in the police department for searching perpetrators quickly and more efficiently as the model will replace the time-consuming and labor-intensive process of traditional forensic search artists. This reflects our motivation to enhance the efficiency of identifying perpetrators involved in a crime activity.

Limitations and Challenges:

The proposed approach of generating facial images from attributes via synthesized sketches opens doors of innovation in the field of generative AI technology. While this technique presents several advantages over the existing models and traditional methods, it also faces certain limitations, which must be addressed to make the model more pragmatic.

First and foremost, the model overlooks the problem of biasness which is the main concern of our project. The fact that the datasets used to train the model are the publicly available CelebA and the LFW datasets which are heavily focused on celebrity faces and do not capture the diversity of facial features in everyday face. Hence, it makes the model more adept at handling facial attributes that are common to celebrities such as high degrees of facial symmetry, clear skin and defined jawline whereas less effective for ordinary faces. Similarly, there is an ethnic imbalance in the dataset as the celebrity database does not contain records from third world developing countries which limits the model's implementation. To mitigate this bias, there is a need to augment the training dataset to make it more diverse and balanced.

Secondly, computational complexity is a significant concern. Processing sketches and generating images requires fast, resource-intensive machines, such as GPUs, to train the model efficiently. While this approach is relatively less expensive than existing direct text-to-facial image generation methods, the stage-wise process does not make it computationally cheap. As mentioned in the paper, the total training time for the proposed model took about 20 hours while using a single NVIDIA Titan X GPU which is the fastest, most advanced graphics card designed primarily for deep learning tasks (Balaban, 2018).

Lastly, the paper lacks analysis for prospects towards implementing the model in a crime investigation agency. This is a fairly difficult task provided that the model may not always match the precision required for real world crime investigation. In addition, the generated sketches or images may not qualify as evidence in courts because they were not drawn by human artist (Technexus, 2023). With the existing bias in the training data for the model, there is a dire need to regularize the use of AI-sketch artist in criminal investigations as it can lead to wrongful convictions.

In conclusion, while the proposed approach for generating facial images offers promising innovations in generative AI, it faces challenges in bias, computational cost, and real-world forensic application. Addressing these limitations is crucial for ensuring its broader and more responsible adoption.

Conclusion

In summation, the research paper provides a comprehensive exploration of the Attribute2Sketch2Face framework, illustrating its innovative approach to generating facial images from visual attributes through a combination of GANs and CVAEs. The detailed methodology and the application of these advanced deep learning models underscore the framework's potential in transforming suspect identification processes. The paper's comparative analysis highlights the framework's superiority in accuracy over existing methods, reinforcing its relevance to our project. However, our analysis also reveals limitations, including potential biases in training data and challenges in computational complexity, of the proposed framework which must be addressed to enhance its practical application. These insights are crucial for refining our own approach and improving the effectiveness of generative AI technologies in forensic and other real-world applications.

Research Paper 2:

Title: A sketch-based image retrieval technique (Abdulaali, Sulong, Abdulbaqi, & Hashem)

Authors: Abdulaali, Soukaena Hassan Hashem, Ghazali Sulong

Main Objective:

The main objective of the paper "A Sketch-Based Image Retrieval Technique" by Abdulaali, Abdulbaqi, Sulong, and Hashem is to develop a robust method for retrieving images based on user-provided sketches. This involves focusing on the extraction of features from sketches to facilitate accurate image matching. The research aims to compare the effectiveness of sketch-based retrieval methods with traditional content-based retrieval approaches. By enhancing the efficiency of mapping sketch features to corresponding images, the technique seeks to improve applications across various fields such as crime investigation, fashion design, and architectural planning. This objective underscores the potential for sketch-based retrieval to provide a more intuitive and efficient way of searching for images in large databases.

Implementation details:

The method for SBIR begins with a user query input in the form of a sketch. This sketch is passed onto the feature extraction stage where the system extracts the relevant features in the sketch such as identifying key characteristics such as shapes, lines and other visual elements that can be used to represent the sketch the way it can be matched against images in the database. Feature extraction techniques include Tensor-Edge Histogram Descriptor (EHD), Sketch Histogram of Gradients Next (ShoG), Scale-Invariant Feature Transform (SIFT) and Bag of Words (BoW). The matching process begins where the extracted features from the sketch are matched against the indexed features of the images

in the database. Moving forward, the system incorporates a relevance feedback mechanism, where users can indicate whether the retrieved images are relevant to their query. This feedback is used to refine future searches and improve the accuracy of the retrieval process. Finally, the system outputs the most relevant images that are closely matching to the input sketch. Figure 5 illustrates the framework discussed in the paper.

Accuracy Measurement and Results

The two main methods for determining the accuracy of the retrieved images are precision and recall, as discussed in the paper. Precision is defined as the ration of number of relevant images retrieved to the total number of images retrieved. On the other hand recall measures the ability of the system to retrieve all relevant images from the database and is defined as the ratio of the number of relevant images retrieved to the total number of relevant images in the dataset. The proposed method employs SIFT (Scale-Invariant Feature Transform) for feature extraction, which has been shown to achieve an impressive accuracy of 95.59%. This high accuracy suggests that the technique is highly effective in accurately retrieving images that match the input sketch, demonstrating the potential of SIFT in enhancing sketch-based image retrieval systems.

Advantages:

Flexibility in Querying: The sketch-based image retrieval technique allows users to input queries in the form of sketches, providing a flexible and intuitive way to search for images, especially when textual descriptions are insufficient or unavailable.

Improved Retrieval Performance: By focusing on extracting and matching visual features, this method improves retrieval performance, enabling more accurate and relevant search results compared to traditional content-based approaches.

Scalability: The technique can be scaled to accommodate large image databases, making it suitable for extensive datasets without significant loss of performance.

User Feedback Mechanism: Incorporating a feedback mechanism allows the system to learn from user interactions, continually refining and enhancing the accuracy of image retrieval.

Sketch as a Universal Language: Sketches are universally understood, transcending language barriers, which makes this retrieval method accessible and useful in a variety of contexts.

Resilience to Scale and Rotation (SIFT): SIFT (Scale-Invariant Feature Transform) is used for feature extraction, which is robust against changes in scale and rotation, ensuring consistent performance even with varied sketch inputs.

Limitations:

Skill Requirement for Effective Sketching: Effective use of the system depends on the user's ability to create accurate sketches. This can be a significant barrier for individuals who lack drawing skills.

Challenges in Feature Extraction and Matching: Extracting and matching features from sketches to images can be complex, particularly when dealing with intricate or abstract drawings.

Dependence on Indexing Structures: The performance of the retrieval system heavily relies on efficient indexing structures to quickly search through large databases, which can be a challenge to implement and maintain.

Poor Performance with Highly Abstract Sketches: The system struggles with highly abstract sketches, as they often lack the detailed features necessary for accurate matching.

Weak Performance in Handling Color: The technique primarily focuses on shape and structural features, resulting in weaker performance when dealing with color-based queries or datasets where color is a significant distinguishing factor.

Conclusion:

In conclusion, the sketch-based image retrieval technique presented in the paper holds significant relevance for Tracefy, an AI system designed to generate accurate criminal sketches from visual attributes. The method's emphasis on feature extraction, flexible querying through sketches, and robust performance with SIFT aligns well with Tracefy's goals of assisting in crime investigations. By leveraging the universal language of sketches and incorporating user feedback, this technique can enhance Tracefy's capability to retrieve and match suspect images accurately. Despite challenges like handling abstract sketches and color, the potential for improved retrieval performance and scalability makes this approach a valuable addition to Tracefy's toolkit, ultimately aiding in the efficient identification of perpetrators.

Research Paper 3:

Title: High-Resolution Image Synthesis with Latent Diffusion Models

Authors: Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, Björn Ommer

Main Objective:

The primary objective of this paper is to propose a new framework for high-resolution image synthesis using Latent Diffusion Models (LDMs). LDMs are designed to tackle the inefficiencies and computational demands of generative models by working within a compressed latent space instead of pixel space, reducing the computational complexity. The authors demonstrate how their approach can generate high-quality images with reduced resource consumption, enabling practical applications in various domains such

as image generation, editing, and more. The paper provides detailed insights into the model architecture, the benefits of latent diffusion, and the results achieved on various datasets, highlighting its potential for real-world applications.

Overview of the Methodology:

The paper introduces Latent Diffusion Models (LDMs), which operate in a latent space to overcome the computational challenges faced by traditional diffusion models working in pixel space. LDMs combine the strengths of latent variable models and diffusion processes for efficient, high-resolution image synthesis.

Latent Space Representation

In the first stage, the model encodes the input image into a compressed latent space. The authors utilize a pre-trained Variational Autoencoder (VAE) to map high-dimensional image data into a low-dimensional latent space. This latent representation captures the essential features of the image while maintaining its overall structure, significantly reducing the computational cost of the subsequent steps. The VAE ensures that complex image features can be represented compactly and efficiently, which is crucial for high-resolution image synthesis.

Diffusion Process in Latent Space

Instead of applying the diffusion process in pixel space, which would be computationally expensive, LDMs apply it in the latent space. A noise-conditional model is trained to denoise the latent representation step by step. Each diffusion step incrementally improves the latent vector's quality by refining its details. The iterative nature of this process allows the model to synthesize high-resolution images with reduced resource requirements, as the latent space is more compact and easier to handle than the pixel space. Figure 6 explains the diffusion process in latent space.

Image Reconstruction

After the diffusion process has refined the latent vector, the model uses a decoder (the second part of the VAE) to map the refined latent space back into the original pixel space. This stage results in a high-resolution image, which closely resembles the target image or adheres to the provided prompts or conditions.

Through this process, LDMs manage to maintain image quality and detail while significantly reducing the computational burden of the generation process.

Implementation and Analysis of Results:

The LDM framework was evaluated on multiple datasets, such as CelebA-HQ, FFHQ, and LSUN. The authors measured the model's performance using metrics like Fréchet Inception Distance (FID) and Inception Score (IS), which are standard for evaluating generative models. They compared LDM's performance with existing methods like GANs, VAEs, and pixel-space diffusion models.

Key Findings:

Efficiency: By operating in latent space, LDMs significantly reduce the computational cost of generating high-resolution images. This makes the model more scalable and feasible for real-world applications compared to traditional diffusion models.

Performance: The LDM approach consistently outperforms pixel-space diffusion models in terms of both speed and quality. The generated images have lower FID scores, indicating higher fidelity to the target images.

Flexibility: LDMs can be adapted to various image synthesis tasks, such as inpainting, super-resolution, and text-to-image generation, further enhancing the versatility of the approach.

Innovation and Contribution:

The authors introduce a novel approach to image synthesis by applying diffusion models in the latent space rather than the pixel space. This key innovation significantly reduces the computational demands of generating high-resolution images, which has been a major bottleneck in previous approaches. The use of VAEs to represent images in latent space is a practical and efficient solution, as it compresses the data while retaining essential features.

Key Contributions:

Efficiency through Latent Space Processing: LDMs reduce the high computational complexity typically required for generating high-resolution images by operating in the latent space.

Improved Image Quality: The model generates images with high detail and realism, as shown by its performance on standard datasets.

Versatile Application: LDMs can be adapted for various image synthesis tasks, such as inpainting and super-resolution, making them applicable to a wide range of real-world scenarios.

Relevance to Our Project (Tracefy):

The paper's approach closely aligns with the goals of our AI-based sketch generation project, particularly in its emphasis on efficiency and image quality. Given that our project aims to synthesize sketches with high precision and speed, the LDM framework can offer several advantages:

Efficiency Gains: The latent space approach used in LDMs can help reduce the computational cost of generating sketches, allowing us to handle larger datasets and produce results more quickly.

High-Quality Image Synthesis: The iterative refinement process in latent space ensures that the generated sketches retain essential structural details, which is critical for accurately representing visual attributes in our sketch generation system.

Scalability: The flexibility of LDMs makes it possible to expand our sketch generation project to include additional features, such as high-resolution sketch refinement or colorization.

By incorporating the principles of LDMs, we can enhance both the performance and scalability of our sketch generation system, ensuring that it meets the practical demands of real-world applications.

Limitations and Challenges:

While Latent Diffusion Models offer several advantages, there are also some limitations to consider:

Complexity of Training: The diffusion process in latent space still requires significant training time, although it is more efficient than pixel-space models.

Dataset Dependency: Similar to other generative models, the performance of LDMs is highly dependent on the quality and diversity of the training data. If the training data lacks variety, the model may struggle to generate high-quality images across a wide range of conditions.

Latent Space Bottleneck: While operating in latent space reduces computational cost, it also introduces a potential bottleneck. If the latent space is too compressed, it may not capture all the nuances of the input image, leading to a loss of fine details.

Conclusion:

The Latent Diffusion Model framework represents a significant advancement in the field of high-resolution image synthesis. By shifting the diffusion process from pixel space to latent space, the authors have introduced a more efficient and scalable method for generating high-quality images. This approach is highly relevant to our AI-based sketch generation project, as it offers a practical solution for improving both performance and scalability. Despite some challenges, such as training complexity and dataset dependency, the LDM framework has the potential to revolutionize various image synthesis applications, including our sketch generation system.

Research Paper 4:

Title: Removing Biased Data to Improve Fairness and Accuracy

Authors: Sahil Verma, Michael Ernst, Rene Just

Main Objective:

This paper's main objective is to address the bias in machine learning models by removing biased data from the training set to improve fairness and accuracy. The authors explore

School of Electrical Engineering and Computer Science, NUST Islamabad, Pakistan

methods to identify and mitigate bias in datasets, leading to more equitable and reliable models. The focus is on techniques that enable fairness while maintaining or even improving model performance. The paper evaluates the impact of these techniques on various datasets, presenting results that demonstrate how bias reduction can enhance model accuracy without compromising fairness.

Overview of the Methodology:

The paper investigates several methods for identifying and mitigating bias in machine learning datasets, including Exploratory Data Analysis (EDA), AIF360 - Sensitive Attributes, Adversarial Debiasing, and Regularization Penalties. The authors use these methods to detect and address biases, leading to improved model performance.

Method 1: Exploratory Data Analysis (EDA)

Exploratory Data Analysis is used to identify patterns and biases within the data. The authors stress the importance of visually inspecting the data before modeling to detect any imbalances or patterns that could introduce bias into the machine learning model. EDA includes analyzing distributions of sensitive attributes like race, gender, and age, to ensure that the dataset is representative of the population.

Method 2: AIF360 - Sensitive Attributes

The authors utilize AIF360, an open-source toolkit developed by IBM, to analyze and mitigate bias related to sensitive attributes such as race, gender, or socioeconomic status. AIF360 provides fairness metrics to evaluate models and datasets for bias, helping researchers identify where models may treat certain groups unfairly. This toolkit also offers debiasing algorithms that adjust the data or model to reduce the unfair impact on specific groups.

Method 3: Adversarial Debiasing

Adversarial Debiasing is a technique where an adversarial network is trained alongside the model to detect and remove biases. The adversary's goal is to identify patterns in the data that the main model might use to make biased predictions. As the adversarial model improves at detecting bias, the main model adjusts its learning process to reduce reliance on biased features. This adversarial training ensures that sensitive attributes do not disproportionately influence the model's decisions.

Method 4: Regularization Penalties

Regularization Penalties are added to the loss function to penalize biased model behaviors. The authors apply regularization techniques to constrain the model's reliance on sensitive attributes, which helps in reducing bias during training. This method ensures that the model's predictions are not unfairly affected by specific attributes, thereby improving fairness without sacrificing accuracy.

Implementation and Analysis of Results:

The authors tested their methods on datasets where fairness and bias are critical factors, such as those used in healthcare, hiring, and criminal justice. They used fairness metrics like Equalized Odds, Demographic Parity, and Disparate Impact to measure the effects of bias removal.

Key Findings:

Increased Fairness: By using methods such as Adversarial Debiasing and AIF360, the authors observed a significant reduction in biased outcomes. Models became less likely to treat individuals unfairly based on sensitive attributes like race or gender.

Maintained or Improved Accuracy: The authors found that removing bias did not necessarily lead to reduced model performance. In many cases, fairness and accuracy improved concurrently, as biased data often introduces noise that impairs predictive power.

Effective Bias Detection: EDA and AIF360 proved effective in detecting biases early in the data pipeline, allowing for timely interventions before training the model.

Innovation and Contribution:

The paper provides a practical approach to reducing bias in machine learning models while maintaining or improving accuracy. By integrating methods like Adversarial Debiasing and Regularization Penalties, the authors demonstrate how fairness and performance can coexist in modern AI systems.

Key Contributions:

Practical Bias Mitigation: The paper provides hands-on methods for practitioners to detect and reduce bias in machine learning models, offering tools like AIF360 and adversarial networks.

Balancing Fairness and Accuracy: The authors emphasize that removing bias does not inherently lead to lower accuracy, challenging the common trade-off narrative between fairness and model performance.

Framework for Bias Detection: By incorporating EDA and AIF360, the authors present a robust framework for detecting and addressing biases at both the dataset and model level.

Relevance to Our Project (Tracefy):

In our AI-based sketch generation project, ensuring fairness and accuracy is critical, especially if the system interacts with sensitive attributes such as gender, race, or age. The methods discussed in this paper can help us reduce any biases that might arise from training data or the model itself.

Bias Detection: We can implement EDA and AIF360 to identify potential biases in our training datasets, ensuring that our sketch generation models are not favoring or discriminating against groups.

Adversarial Debiasing: Integrating adversarial debiasing techniques can further ensure that our sketch generation system is fair and unbiased, making the system more reliable and ethical.

Accuracy Maintenance: The paper's focus on maintaining accuracy while improving fairness is directly applicable to our project, as we strive to produce high-quality sketches without compromising ethical considerations.

Conclusion:

The techniques presented in this paper offer practical solutions for addressing bias in machine learning models, with a focus on fairness and accuracy. By using methods such as EDA, AIF360, Adversarial Debiasing, and Regularization Penalties, practitioners can identify and reduce bias while maintaining model performance. For our AI-based sketch generation project, these techniques can be directly applied to ensure that our system generates fair and accurate results across diverse user groups. Despite some challenges, the methods presented provide a solid foundation for creating ethical and high-performing AI systems.

Existing System:

Title: Forensic Sketch AI-rtist

Authors: Artur Fortunato · Filipe Reynau

Overview of the System

Text-to-Facial Image Generation: The Spanish developers' system takes textual descriptions and generates corresponding facial images using DALL-E. This model is pre-trained on a diverse dataset to understand and generate various visual concepts from text.

Bias and Fairness Concerns: Despite DALL-E's advanced capabilities, there are significant concerns regarding bias. The training data for DALL-E, like many large language models, may include inherent biases related to race, gender, and other demographic factors. As a result, the generated images might reflect or amplify these biases.

Limitations of the System

Bias in Generated Images: The primary concern is that the facial images generated by DALL-E might perpetuate or even exacerbate existing biases. For instance, if the training data underrepresents certain demographics or includes stereotypical associations, the system could produce biased or inaccurate representations.

Lack of Contextual Understanding: While DALL-E is powerful, it may struggle with nuanced or complex descriptions, leading to images that might not fully align with the intended

textual description. This could be problematic in applications requiring high precision, such as forensic sketching.

Ethical and Fairness Issues: The system's reliance on pre-trained models raises ethical questions about fairness. Ensuring that generated images do not unfairly represent or misrepresent individuals based on race, gender, or other factors is challenging but crucial.

Tracefy Comparison

Tracefy focuses on generating accurate criminal sketches from visual attributes to assist in identifying suspects. It incorporates Latent Diffusion Models (LDMs) and GANs, with a strong emphasis on fairness and bias reduction techniques.

Fairness and Bias Reduction: Tracefy integrates techniques like Adversarial Debiasing and Regularization Penalties to mitigate bias, aiming to create sketches that are fair and representative. This contrasts with the potential biases inherent in systems like DALL-E.

Specific Use Case: Tracefy's application is more targeted towards forensic investigations, which demands high accuracy and fairness in sketch generation. Unlike DALL-E, which is designed for broader image generation tasks, Tracefy's focus on criminal sketches requires careful consideration of biases and ethical implications in a sensitive context.

Ethical Considerations: By addressing fairness and bias reduction explicitly, Tracefy aims to avoid unintended biases that could impact the effectiveness and ethicality of the sketches used in law enforcement.

In summary, while the Spanish developers' system utilizes DALL-E to generate facial images from text, it faces significant challenges related to bias and fairness. Tracefy, on the other hand, addresses these challenges head-on with specific techniques and considerations tailored to its forensic application, ensuring that the generated sketches are both accurate and unbiased.

Project Overview

Our project focuses on developing an AI-based sketch generation system that automatically creates detailed sketches based on user inputs. The system utilizes generative AI models such as Latent Diffusion Models (LDMs) and Generative Adversarial Networks (GANs), inspired by frameworks like Attribute2Sketch2Face (Di & Patel, 2017), to generate high-quality and accurate facial sketches from visual descriptions. Additionally, fairness and bias reduction techniques such as Adversarial Debiasing and Regularization Penalties are implemented to ensure that the generated sketches do not contain unintended biases related to sensitive attributes like race, gender, or age. This technology aims to assist law enforcement agencies in efficiently identifying suspects based on witness accounts or investigative leads.

The project aims to provide a tool that is both accurate and ethical, helping law enforcement agencies improve the efficiency and effectiveness of criminal investigations, while maintaining fairness in sketch generation across diverse populations.

Key components of the project include:

Data Collection and Preprocessing: Gathering and preparing datasets, including human sketches and reference images, to train the model. **Attribute2Sketch2Face:** Implementing this approach but using LDMs instead of GANs.

Latent Diffusion Models (LDMs): Leveraging diffusion models to generate high-resolution, accurate sketches.

Bias Detection and Mitigation: Implementing methods such as Exploratory Data Analysis (EDA), AIF360, and Adversarial Debiasing to reduce bias in the model outputs.

Model Evaluation: Using fairness metrics and performance indicators to assess the accuracy and fairness of the generated sketches.

Overall, the project combines cutting-edge AI methodologies with fairness principles to deliver a powerful and ethical sketch generation tool for real-world applications.

Project Development Methodology

Our project's development methodology follows a systematic approach to generate facial images from visual attributes using Latent Diffusion Models (LDMs). The key stages of this process are outlined below:

1. Attribute-to-Sketch (A2S) using LDMs

In the first stage, a rough sketch is generated based on given visual attributes. This phase utilizes LDMs, which are particularly effective for modeling complex distributions, such as visual attributes, and progressively refining the quality of synthesized images. Unlike GANs, LDMs introduce noise to the latent space and iteratively reverse this noise, enhancing the sharpness and detail of the generated sketch.

LDM Process: The model learns latent representations of sketches conditioned on facial attributes. It then progressively denoises the representation to generate a detailed sketch.

2. Sketch-to-Sketch (S2S) Refinement

Once the rough sketch is obtained, we move to the refinement stage, where the generated sketch is sharpened to a more realistic and higher-quality image. Again, LDMs are applied here to enhance the finer details of the sketch by denoising the latent representation.

iteratively. This allows the model to capture subtle variations in facial features, resulting in a refined sketch output.

3. Sketch-to-Face (S2F) Transformation

In the final stage, the high-quality sketch from the second stage is used to generate a realistic facial image. This step involves the application of LDMs to transform the sketch into a full-color facial image by incorporating texture, color, and other visual attributes. The diffusion process ensures that the generated face is consistent with the input sketch and attributes, producing high-quality, photorealistic results.

Implementation Tools and Libraries:

Latent Diffusion Models (LDMs) for sketch generation and refinement.

Python with libraries such as PyTorch for model training and implementation.

Data Sources: Facial image datasets like CelebA and CUHK Face Sketch Database.

Application framework: The application will be made in Flask with a user-friendly UI in React

Relevance to UN Sustainable Development Goals

The United Nations Sustainable Development Goals are a set of instructions for countries around the world to devise strategies that improve health and education, reduce inequality and spur economic growth while tackling climate change and working to preserve natural resources (UN Sustainable Development Goals, n.d.). Keeping these in mind we are looking forward to mapping the outcomes of our project to six sustainable development goals which are mentioned further in the order of relevance.

First and foremost, Tracefy aligns with SDG 16 (Peace, Justice, and Strong Institutions) by enhancing law enforcement capabilities hence promoting accountability in the justice system and making it a lot quicker than it is right now due to the traditional methods of sketching a perpetrator. The system also supports SDG 11 (Sustainable Cities and Communities) by contributing to safer urban environments and improving the standard of living in major cities of Pakistan. By leveraging advanced AI technologies, Tracefy promotes SDG 9 (Industry, Innovation, and Infrastructure), encouraging innovation and the development of robust legal infrastructures and providing an opportunity for us to work on the implementation of this unique idea. Additionally, we are hopeful that Tracefy will be a valuable contribution towards achieving SDG 8 (Decent Work and Economic Growth) by potentially creating new jobs in tech and security sectors. Through its inclusive approach to reduce bias, Tracefy supports SDG 10 (Reduced Inequalities) by reducing the number of wrongful accusations against marginalized communities. Moreover, we look forward to its

implantation in rural areas where infrastructure for ensuring peace is not sufficient to identify miscreants involved in a situation. Lastly, we will make sure that Tracefy is easily applicable to real life scenarios hence it will indirectly contribute to SDG 3 (Good Health and Well-being) by reducing crime, thus enhancing community well-being and safety.

By integrating AI technology into criminal sketching, Tracefy is expected to not only address critical justice and security needs but also foster innovation and economic growth while promoting social equity. We are confident that Tracefy will significantly contribute to these Sustainable Development Goals, creating a safer and more just society for all.

Work Division

Mahad: Application Structure

Designs the architecture of the application.

Ensures the app has a solid foundation for efficient functionality.

Mehar: Accurate Prediction Search

Searches for precise algorithms and techniques.

Focuses on enhancing the application's prediction performance.

Rafay: Fine-tuning

Refines and optimizes the model's performance after initial implementation.

Adjusts parameters and features for improved accuracy and efficiency.

Shared Responsibilities

Data collection, model selection, and implementation.

Each member contributes to:

Gathering necessary data.

Selecting Latent Diffusion Models (LDMs).

Ensuring smooth integration into the application workflow.

Timeline

Data Finalization (8 weeks)

Completion of data collection and preparation.

Ensures that the dataset is ready for model training.

Model Training and Fine-tuning (16 weeks)

Training the selected Latent Diffusion Models (LDMs).

Fine-tuning the model to enhance its accuracy and performance.

Search Function (8 weeks)

Development and integration of the search functionality.

Ensures accurate and efficient search results based on user input.

Application Structure (4 weeks)

Finalization of the application's architecture.

Completion of the overall structure for seamless functionality and user experience.

Costing

The pie chart presents a rough estimation of the cost distribution for implementing a sketch-based image retrieval system. The cost is divided into five main categories:

API Keys (35%):

A significant portion of the budget is allocated to API keys, which are essential for accessing various third-party services and datasets required for the system. These could include APIs for image processing, machine learning frameworks, or databases.

GPU Requirements (30%):

Nearly a third of the cost is dedicated to GPU requirements. High-performance GPUs are necessary for training and running deep learning models efficiently, which are crucial for tasks like feature extraction and image matching.

Data Annotation (20%):

Data annotation constitutes 20% of the total cost. Annotating data involves labeling sketches and corresponding images, which is a labor-intensive process but essential for training accurate models.

Hosting (10%):

Hosting costs account for 10% of the budget. This includes the expenses for cloud services or physical servers where the system will be deployed and maintained.

Miscellaneous (5%):

The remaining 5% is reserved for miscellaneous expenses. These might include unforeseen costs, minor software purchases, or additional resources needed during the development and deployment phases.

This cost distribution provides a clear overview of where many of the resources will be invested, highlighting the emphasis on essential components like API access, GPU capabilities, and data preparation.

References:

Crime Index for Karachi and Rawalpindi (2024) Crime Index – Pakistan

Ayaz, D. (2023, February 11). Combatting Street Crimes in Pakistan. Retrieved from Paradigmshift: <https://www.paradigmshift.com.pk/street-crimes-pakistan/>

Xing Di · Vishal M. Patel (2017) Face Synthesis from Visual Attributes via Sketch using Conditional VAEs and GANs <https://doi.org/10.48550/arXiv.1801.00077>

Abdulaali · Abdulbaqi · Sulong · Hashem (2014) A sketch-based image retrieval technique
<https://www.researchgate.net/publication/305650630>

Artur Fortunato · Filipe Reynau (2022) Forensic Sketch AI-rtist
<https://lablab.ai/event/openai-whisper-gpt3-codex-dalle2-hackathon/eagleai/forensic-sketch-airtist>

R Rombach · A Blattmann · D Lorenz · P Esser · Björn Ommer (2021) High-Resolution Image Synthesis with Latent Diffusion Models <https://doi.org/10.48550/arXiv.2112.10752>

Sahil Verma · Michael Ernst · Rene Just (2021) Removing biased data to improve fairness and accuracy <https://doi.org/10.48550/arXiv.2102.03054>

G4IE & MPII - Google Drive

faces-sketch-dataset – Kaggle

Person Face Sketches – Kaggle

CUHK Face Sketch Database (CUFS) – Kaggle

Arrorra, R. (2019). ResNets, DenseNets & UNets <https://medium.com/swlh/resnets-densenets-unets-6bbdbcfdf010>

Balaban, M. (2018) Titan RTX Benchmark <https://lambdalabs.com/blog/titan-rtx-tensorflow-benchmarks>

Rosenberg, H., & Ahmed, S. (2023) Limitations of Face Image Generation
<https://arxiv.org/html/2309.07277v2>

Technexus. (2023). Developers Created AI to Generate Police Sketches
<https://medium.com/@Technexus.ai/developers-created-ai-to-generate-police-sketches-58f85131d125>

Latent Diffusion Model - Hugging Face <https://huggingface.co/CompVis/ldm-text2im-large-256>

Appendix:



Figure 1: Results of Stage 1

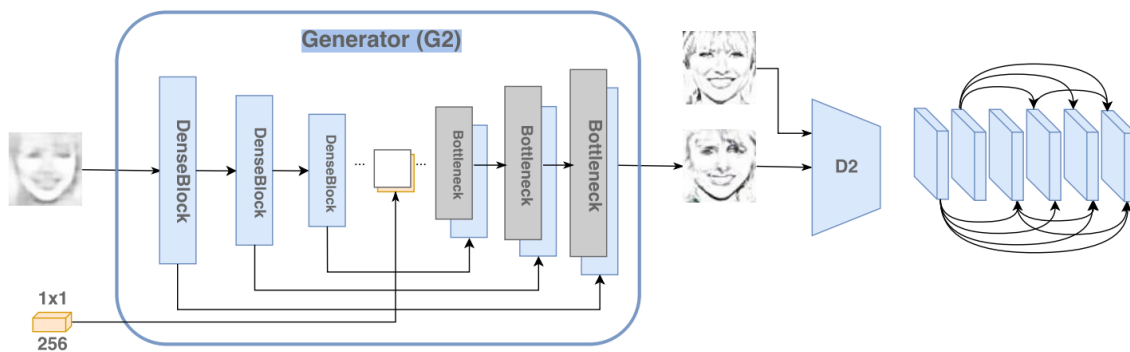


Figure 2: Workflow of Stage 2

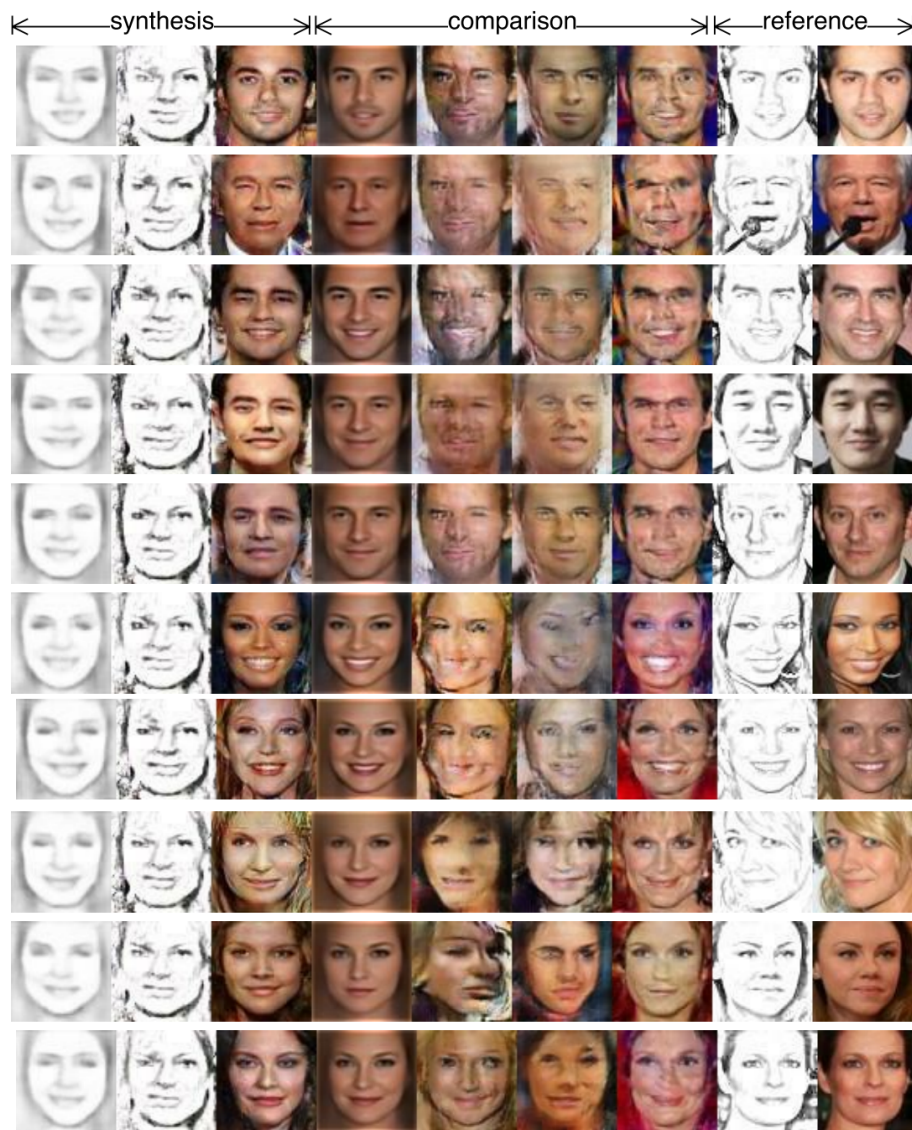


Figure 3: Results of the experiment

Metric	Dataset	text2img [31]	StackGAN [41]	CVAE [40]	attr2face	Attribute2Sketch2Face
Inception Score	CelebA	1.486 ± 0.016	1.517 ± 0.014	1.275 ± 0.005	1.524 ± 0.010	1.545 ± 0.011
	LFW	1.510 ± 0.020	1.589 ± 0.018	1.482 ± 0.017	1.592 ± 0.081	1.617 ± 0.025
	CUHK	1.269 ± 0.049	1.349 ± 0.114	1.119 ± 0.027	1.345 ± 0.131	1.417 ± 0.040
Attribute L_2	CelebA	0.104 ± 0.024	0.091 ± 0.021	0.080 ± 0.042	0.081 ± 0.023	0.079 ± 0.022
	LFW	0.093 ± 0.027	0.085 ± 0.029	0.086 ± 0.019	0.073 ± 0.035	0.069 ± 0.034
	CUHK	0.610 ± 0.018	0.063 ± 0.019	0.064 ± 0.020	0.062 ± 0.020	0.058 ± 0.021

Figure 4: Quantitative Analysis

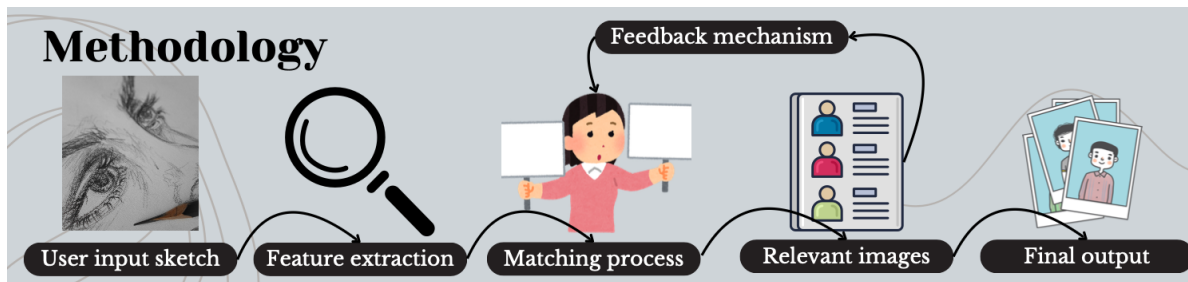


Figure 5: Sketch Based Image Retrieval Mechanism

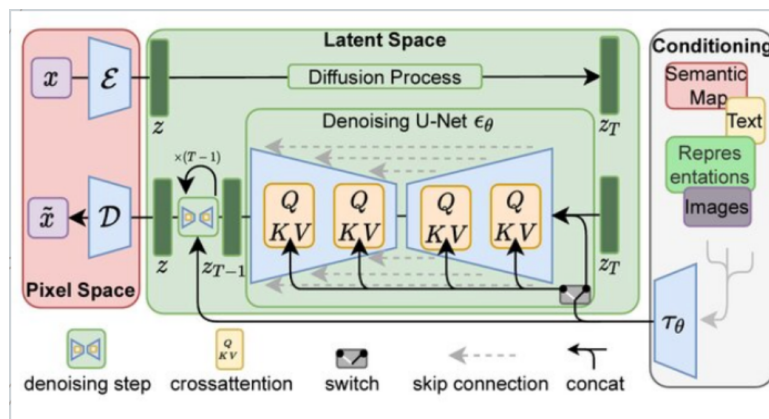


Figure 6: Latent Diffusion Models Workflow